

ST GALLEN, Switzerland

WMO: 066810

Lat: 47.4256N

Lon: 9.3986E

Elev: 2552

StdP: 13.39

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 14.0	(c) 17.4	(d) 9.1	(e) 9.6	(f) 19.9	(g) 12.3	(h) 11.3	(i) 21.5	(j) 20.7	(k) 43.4	(l) 18.4	(m) 42.4	(n) 3.0	(o) 30	(p) 0.495

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 13.4	(c) 80.7	(d) 65.4	(e) 77.8	(f) 64.1	(g) 75.2	(h) 63.1	(i) 67.4	(j) 77.4	(k) 65.7	(l) 75.2	(m) 64.3	(n) 73.3	(o) 5.6	(p) 30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.6	96.9	73.6	62.0	91.5	71.5	60.6	86.7	69.4	33.4	77.3	32.0	75.3	30.8	73.3	75.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.3	13.4	11.3	10.6	85.3	4.6	2.8	7.3	87.3	4.6	88.9	2.0	90.5	-1.3	92.5
			9.6	71.0	4.5	1.9	6.3	72.3	3.7	73.4	1.1	74.4	-2.2	75.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	48.1	32.2	33.7	40.2	47.0	54.6	60.9	64.1	63.4	56.0	49.2	40.0	34.3	(6)
(7)		DBStd	13.67	8.51	9.06	8.28	8.14	7.24	7.36	6.32	6.27	6.40	7.38	8.11	8.31	(7)
(8)		HDD50	2439	552	456	315	150	36	4	0	1	18	105	314	488	(8)
(9)		HDD65	6366	1016	875	769	539	332	165	95	107	276	489	751	952	(9)
(10)		CDD50	1728	0	1	12	61	177	332	437	416	198	81	12	1	(10)
(11)		CDD65	181	0	0	0	0	9	42	66	58	6	0	0	0	(11)
(12)		CDH74	829	0	0	0	1	29	213	333	245	8	0	0	0	(12)
(13)		CDH80	117	0	0	0	0	1	28	51	37	0	0	0	0	(13)
(14)	Wind	WSAvg	4.4	4.4	4.5	5.1	5.0	4.9	4.4	4.4	4.1	3.9	3.7	4.1	4.5	(14)
(15)	Precipitation	PrecAvg	57.5	3.2	3.5	3.9	4.1	5.8	6.7	6.8	6.5	4.9	4.0	4.3	4.0	(15)
(16)		PrecMax	74.4	7.2	9.6	8.2	7.3	13.2	10.6	12.8	10.9	8.3	8.1	9.3	9.1	(16)
(17)		PrecMin	45.0	0.3	1.1	1.4	1.1	1.2	2.8	2.8	1.5	2.3	0.5	0.2	0.4	(17)
(18)		PrecStd	6.1	1.9	2.0	1.7	1.7	2.6	2.0	2.4	2.4	1.5	1.7	2.0	2.0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.8	56.0	62.0	70.3	77.3	83.6	85.0	84.2	74.9	69.5	64.1	58.4	(19)
(20)			MCWB	43.6	43.6	48.7	53.9	60.5	66.0	67.4	66.5	62.1	55.2	49.9	45.6	(20)
(21)		2%	DB	50.2	51.9	58.0	66.3	73.5	79.4	80.8	79.7	71.6	65.0	59.3	51.7	(21)
(22)			MCWB	42.7	41.9	46.6	51.9	58.7	64.4	65.1	66.0	61.4	55.3	48.3	43.7	(22)
(23)		5%	DB	47.0	48.9	55.0	63.1	69.7	76.3	77.6	76.4	68.6	62.1	54.8	48.7	(23)
(24)			MCWB	40.8	40.5	45.2	50.5	56.8	63.3	64.3	64.5	59.7	54.1	47.5	42.4	(24)
(25)		10%	DB	44.0	46.0	52.1	59.7	66.4	72.9	74.7	73.6	65.7	59.3	51.3	45.9	(25)
(26)			MCWB	38.8	39.1	43.9	48.9	55.2	61.9	63.2	63.2	58.4	52.9	45.7	40.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.6	45.6	50.7	55.8	63.1	69.3	69.7	69.6	64.9	59.5	52.2	46.8	(27)
(28)			MCDWB	50.9	52.1	58.8	66.9	73.8	80.0	80.7	79.6	72.3	64.7	59.7	52.5	(28)
(29)		2%	WB	43.7	43.4	48.3	53.5	60.4	66.4	67.2	67.3	62.7	57.1	49.9	45.1	(29)
(30)			MCDWB	48.5	49.9	55.5	63.9	70.6	76.2	77.5	77.0	69.5	62.3	55.8	50.7	(30)
(31)		5%	WB	41.4	41.6	46.3	51.5	58.1	64.4	65.5	65.6	60.9	55.3	48.0	43.2	(31)
(32)			MCDWB	46.4	47.4	53.1	61.1	67.4	74.0	75.5	74.6	67.1	60.5	53.9	48.0	(32)
(33)		10%	WB	39.2	39.6	44.6	49.5	56.1	62.6	63.8	64.0	59.1	53.6	46.1	41.2	(33)
(34)			MCDWB	43.6	45.1	51.1	57.9	64.7	71.4	73.2	72.1	64.3	58.4	51.0	45.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.8	8.9	11.2	13.1	13.5	13.3	13.4	12.5	11.2	9.8	8.6	8.4	(35)
(36)			MCDBR	11.3	12.8	15.1	17.1	17.6	17.4	17.0	16.4	14.7	13.5	14.1	12.3	(36)
(37)			MCWBR	7.5	7.7	8.9	9.6	9.9	10.0	9.7	9.8	8.9	8.5	7.7	7.5	(37)
(38)		5% WB	MCDWB	10.4	11.7	13.5	16.0	16.3	16.0	15.8	15.2	13.4	11.6	12.1	11.2	(38)
(39)			MCWBR	7.6	7.9	8.6	9.5	9.6	9.6	9.4	9.6	8.6	8.2	7.5	7.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.275	0.298	0.335	0.371	0.385	0.392	0.393	0.385	0.358	0.330	0.300	0.271	(40)	
(41)		taud	2.431	2.371	2.358	2.309	2.314	2.331	2.320	2.358	2.417	2.484	2.507	2.448	(41)	
(42)		Ebn at Noon	261	276	281	279	279	277	274	272	270	261	249	248	(42)	
(43)		Edh at Noon	21	28	33	38	39	39	39	36	31	25	20	18	(43)	
(44)	All-Sky Solar Radiation	RadAvg	371	639	1022	1375	1581	1815	1756	1533	1150	724	404	305	(44)	
(45)		RadStd	41	86	137	198	159	169	160	157	118	79	49	47	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	N/A	+1.22	+0.77	+0.65	N/A	N/A	+151	+47	
(47)	Regional (115 neighbors)	N/A	N/A	N/A	+1.21	+0.76	+0.65	N/A	N/A	+97	+23	(47)

Nomenclature: See separate page